Polish-Japanese Academy of Information Technology

Bachelor's engineering thesis

Short-Term Forecasting of Martian Surface Weather Using Machine Learning

PRO3, Task 03: Chapter 3, Data and materials

Uladzislau Staliarou (s29905)

October 2026

3 Data and materials

This chapter documents the data on which the thesis rests: their sources, formats and metadata, their storage and naming, the rules by which they were cleaned, the meaning of every variable in the derived tables and the terms under which they may be used. The exploratory analysis of the data and their standardization and normalization follow in Chapter 4, and the software that processes them is described in Chapter 5.

3.1 Data types and sources

All inputs and targets of the forecasting models are tabular time series, with one row per measurement or time step and one column per variable. Text enters the project in three forms, none of which is analyzed as data: the metadata of the archives (labels, format files, readme files and release notes), the configuration files of the experiments and the source code. The only images are the figures produced for this thesis. The MEDA archive also holds sky images from the SkyCam camera, but they are not used, and the project contains no audio or video data. The rest of this chapter therefore concerns tabular data and their textual metadata. All physical quantities are stored in SI units (kelvin, pascal, meters per second, watts per square meter), except the water-vapor mixing ratio, given in parts per million, and angles, given in degrees.

The data come from three public archives of the NASA Planetary Data System (PDS), all distributed by its Planetary Atmospheres Node at New Mexico State University. Table 3.1 lists them with the digital object identifiers (DOIs) under which they are cited. Each archive offers the records of one station at several processing levels. MEDA, for example, publishes raw, partially processed, calibrated and derived data. Calibrated data are the lowest level in physical units, and derived data add retrievals and models on top of them. The MEDA team recommends the derived level wherever it exists and the calibrated level otherwise. It also warns that where a derived product has empty columns, the data of that interval are not reliable and the calibrated values must not be used in their place[1]. This thesis applies the same rule to all three stations and uses the highest processing level available for each variable.

[1] J. A. Rodriguez-Manfredi et al. *The Mars Environmental Dynamics Analyzer, MEDA*. Data bundles of the Mars 2020 mission. NASA Planetary Data System, Atmospheres Node. 2021. URL: `https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/PERSEVERANCE/meda.html` (accessed 2026-10-07). DOI: 10.17189/1522849, `readme.txt` of the bundle.

Table 3.1: Archives used in this thesis

Station (site)	Data set and DOI	Processing level and products	Sols used (UTC period)	Files and size
MEDA (Jezero Crater)	Mars 2020 MEDA bundle, release 14; 10.17189/1522849	derived: PS, WS, RHS, TIRS, ANCILLARY; calibrated: ATS, RDS	1–1619 (19 Feb 2021 to 9 Sep 2025)	9,198 CSV files, 41 GB
REMS (Gale Crater)	MSL-M-REMS-5-MODRDR-V1.0; 10.17189/ghbc-qr41	MODRDR: air temperature, pressure, wind	4,703 sols between 1 and 4843 (7 Aug 2012 to 22 Mar 2026)	4,703 tables, 50 GB
APSS (Elysium Planitia)	APSS TWINS bundle; 10.17189/1518950	derived (modeled): air temperature, wind	747 sols between 4 and 1333 (30 Nov 2018 to 27 Aug 2022)	747 CSV files, 5.8 GB
	APSS PS bundle; 10.17189/1518939	calibrated: pressure	926 sols between 4 and 1218 (30 Nov 2018 to 30 Apr 2022)	926 CSV files, 55 GB

Source: own elaboration based on the PDS archives[2].

For MEDA, derived products exist for pressure (PS), wind (WS), humidity (RHS), the thermal infrared sensor (TIRS) and the ancillary geometry (ANCILLARY), whereas air temperature (ATS) and radiation (RDS) are released only at the calibrated level. The release used is the fourteenth, published in December 2025, which ends with sol 1619[3]. For REMS, the thesis uses the MODRDR data set, whose tables hold the environmental magnitudes "after applying filters, corrections and models" to the lower-level records, "thus containing the best estimation of the real data"[4]. Air temperature and wind at the InSight site come from the derived, so-called modeled, collection of the TWINS bundle[5], and pressure from the calibrated collection of the PS bundle[6].

The stations sample the atmosphere differently, which matters whenever their records are compared. REMS records five minutes of every hour at 1 Hz, with additional blocks of one full hour[7]. MEDA records one-hour sessions that alternate between even and odd hours[8]. In the files

[2] *Idem*, *MEDA data bundles*; J. Gomez-Elvira. *Mars Science Laboratory Rover Environmental Monitoring Station RDR Data V1.0*. MSL-M-REMS-5-MODRDR-V1.0. NASA Planetary Data System, 2013. URL: https://pds.nasa.gov/ds-view/pds/viewDataset.jsp?dsid=MSL-M-REMS-5-MODRDR-V1.0 (accessed 2026-10-18); J. A. Rodriguez-Manfredi. *APSS TWINS Data*. NASA Planetary Data System, 2019. URL: https://pds.nasa.gov/ds-view/pds/viewBundle.jsp?identifier=urn:nasa:pds:insight_twins (accessed 2026-10-18); D. Banfield. *APSS PS Data*. NASA Planetary Data System, 2019. URL: https://pds.nasa.gov/ds-view/pds/viewBundle.jsp?identifier=urn:nasa:pds:insight_ps (accessed 2026-10-18).

[3] Rodriguez-Manfredi et al., *op. cit.*, release notes, release 14.

[4] Gomez-Elvira, *op. cit.*, label of each MODRDR product.

[5] Rodriguez-Manfredi, *op. cit.*

[6] Banfield, *op. cit.*

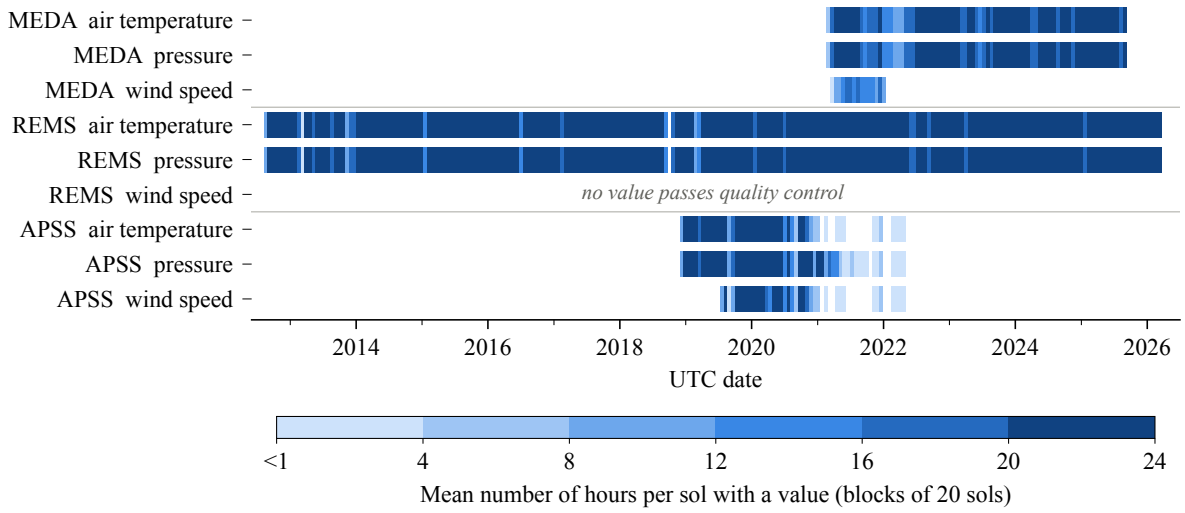
[7] G. M. Martínez et al. "The Modern Near-Surface Martian Climate: A Review of In-situ Meteorological Data from Viking to Curiosity". In: *Space Science Reviews* 212.1–2 (2017), pp. 295–338.

[8] J. A. Rodriguez-Manfredi et al. "The diverse meteorology of Jezero crater over the first 250 sols of Perseverance

used here, the remaining hours usually contain a sample of about five minutes, and from about sol 1200 onwards MEDA recorded about half of most hours. APSS recorded almost continuously, wind and air temperature at up to 1 Hz and pressure at up to 20 Hz[9]. An hourly mean therefore rests on about five minutes of data at Gale, on five minutes to a full hour at Jezero and usually on a full hour at Elysium Planitia (Section 3.8).

The data were downloaded over HTTP from the archive of the node, MEDA in April 2026 and REMS and InSight in October 2026. A download script mirrors the archive folders, resumes interrupted transfers and checks file sizes (Section 3.8). For REMS, the list of files up to sol 4133 was taken from the index table of the archive volume. The products of sols 4000–4133 had been re-released as newer versions after the index was written, and later sols were not yet indexed, so these folders were read from the directory listings instead. Six of the 932 daily InSight pressure files (sols 1148, 1195, 1197, 1200, 1217 and 1219) could not be downloaded over an unstable connection. They fall in the sparse late phase of the mission described below.

Figure 3.1: Temporal coverage of the station records used in this thesis



Source: own elaboration based on the hourly tables described in Section 3.5.

Figure 3.1 shows when each record holds usable values. Three features of the figure matter for the later chapters. First, the records overlap little in time. Only REMS spans the whole period, and the InSight pressure record ends in April 2022, about fourteen months after Perseverance landed. Second, wind is the weakest variable at every site. Derived MEDA wind exists only for sols 36–312, because parts of the wind sensor were damaged by the harsh weather and its recalibration was still in progress[10]. No REMS wind sample passes the quality control (Section 3.4), and InSight wind is missing from sol 20 to about sol 220, when it was sampled at 0.1 to 0.5 Hz. Third, the InSight record thins out late in the mission. Up to about sol 800, the

on Mars". In: *Nature Geoscience* 16 (2023), pp. 19–28.

[9]D. Banfield et al. "The atmosphere of Mars as observed by InSight". In: *Nature Geoscience* 13.3 (2020), pp. 190–198.

[10]Rodriguez-Manfredi et al., *MEDA data bundles*, release notes, section B.4.

median sol holds all 24 hourly values of air temperature and of pressure. After sol 870, only 71 sols have any air temperature and 138 any pressure, with a median of six and five hourly values. The usable InSight record for the transfer experiments therefore spans about sols 4–870.

3.2 Formats, metadata and naming conventions

Table 3.2 summarizes how the three archives store their data. MEDA and InSight follow the current PDS4 standard. Every data file is a comma-separated values (CSV) table with a header row, holds one sol of one sensor or product and is accompanied by an Extensible Markup Language (XML) label. The label gives the name, data type, unit and a short description of every column, so the files are self-describing and their columns can be read by name. REMS follows the older PDS3 standard. Its tables are fixed-width text files with comma-separated fields but no header row. Each table has a plain-text label that points to a separate format file, which defines the 40 columns by their position[11]. The pipeline therefore reads REMS columns by position, as defined in that file.

Table 3.2: Formats, metadata and naming conventions of the archives

Property	MEDA	REMS	APSS
Standard	PDS4	PDS3	PDS4
Data file	CSV with header row; one file per sol and sensor	fixed-width text, comma-separated, no header row; one table per sol	CSV with header row; one file per sol and product
Metadata	XML label per file	text label per file; columns defined in a format file	XML label per file
File name	prefix `WE`, sol, level, sensor, version	prefix `RME`, clock count, product type, sol, version	product, sol, version
Folders	blocks of sols, then one folder per sol	blocks of sols, then one folder per sol	blocks of sols
Time fields	SCLK, LMST, LTST	SCLK (named TIMESTAMP), LMST, LTST	onboard time, SCLK, LMST, LTST, UTC
Missing values	empty cell; 999999999	−999	empty cell
Quality information	empty columns of derived products; rover-state flags	confidence strings, one character per check	operational flag strings, one character per check
Versions on disk	P01–P04	P1–P9, PA	01–03

Source: own elaboration based on the labels and documentation of the archives.

File names encode the content of each file. A MEDA name such as `WE__0300_____ ___DER_PS_____P01.CSV` contains the prefix of the environmental sensors (`WE`), the sol (0300), the processing level (`DER` for derived, `CAL` for calibrated), the sensor (`PS`)

[11] Gomez-Elvira, *op. cit.*, format file `MODRDR6.FMT`.

and the product version (P01). A REMS name such as RME_492079991RMD1066000000 0_____P2.TAB contains the spacecraft clock count at the start of the product, the product type (RMD for MODRDR), the sol (1066) and the version (P2). InSight names are shorter: twins_model_0331_01.csv is the modeled TWINS product of sol 331 in version 1, and ps_calib_0331_01.csv is the calibrated pressure product of the same sol.

The archives version their products. When data are recovered later or the processing is corrected, a new version of a file is published, and only the highest version should be used[12]. The files on disk span several versions (Table 3.2), but in each archive exactly one version is present for every sol and sensor, so no choice between versions is needed. For REMS, this required the directory listings mentioned above, because the index of the volume still pointed to the replaced first versions of sols 4000–4133.

Time is recorded in several forms. All files carry the spacecraft clock (SCLK), a count of seconds, which REMS counts from noon of 1 January 2000[13]. All files also carry local mean solar time (LMST) and local true solar time (LTST) as text. An LMST value such as 00300M02:00:30.032 joins the sol number to a clock time on a 24-hour Mars clock, whose hour is one twenty-fourth of a sol of 24 h 39 min 35.244 s[14]. Only the InSight files also give Coordinated Universal Time (UTC). The pipeline uses LMST to place each sample in its sol and hour, and it estimates UTC as the spacecraft clock plus an epoch fixed on one product of each mission, which is accurate to about one minute. From UTC, the areocentric solar longitude L_s, the angle that measures the Martian season, is computed with the Mars24 algorithm of Allison and McEwen[15]. Each mission counts sols from its own landing, so sol numbers are comparable only within one station.

Missing values are marked differently in each archive (Table 3.2). MEDA leaves cells empty and writes 999999999 where a value could not be computed, REMS writes −999, and InSight leaves cells empty. The MEDA readme gives the invalid value as 9999999999, with one more digit than the value found in the files; the pipeline converts the value that the files contain. Quality information is also stored as text. REMS and InSight attach to each sample strings of flag characters, in which every position answers one question, for example whether the rover was still or whether the electronics were within their temperature range, typically with 1 for good, 0 for bad and X for unknown. These strings are read as text rather than numbers, so that their leading characters and the X marks are preserved.

The labels also revealed a misleading column name. The MEDA ancillary column SOLAR_LONGITUDE_ANGLE does not contain the solar longitude L_s. According to its label, it is the azimuth of the Sun in the local frame of the rover, which runs from −180° to 180° within

[12] Rodriguez-Manfredi et al., *op. cit.*, readme.txt of the bundle.

[13] Gomez-Elvira, *loc. cit.*

[14] Rodriguez-Manfredi et al., *loc. cit.*

[15] M. Allison and M. McEwen. “A Post-Pathfinder Evaluation of Areocentric Solar Coordinates with Improved Timing Recipes for Mars Seasonal/Diurnal Climate Studies”. In: *Planetary and Space Science* 48.2–3 (Feb. 2000), pp. 215–235.

every sol[16]. The pipeline never uses it as a seasonal variable (Section 3.5).

3.3 Folder structure and versioning

Table 3.3 shows how the data are organized. Raw and derived data are kept apart. The three archive folders are mirrors of the PDS archives that keep the original directory layout and file names, so every file can be traced to its source and the labels and index tables remain valid. No processing step writes into these folders. They are treated as read-only, and only the download script adds files to them. Everything derived from them is written to two other folders, `data/interim/` and `data/processed/`, and can be rebuilt from the raw folders with the scripts listed in Section 3.8.

Table 3.3: Folder structure of the data and of the run records

Folder or file	Contents	Format	Written by
`data/meda/meda_derived/`	MEDA derived products (PS, WS, RHS, TIRS, ANCILLARY), readme, release notes, documents	CSV, XML, TXT, PDF	download; read-only
`data/meda/calibrated_subset/`	MEDA calibrated products (ATS, RDS)	CSV, XML	download; read-only
`data/rems/mslrem_1001/`	REMS MODRDR tables and labels, volume index, format file	TAB, LBL, FMT	download; read-only
`data/twins/twins_bundle/`	InSight TWINS modeled products	CSV, XML	download; read-only
`data/twins/ps_bundle/`	InSight PS calibrated products, list of missing files	CSV, XML, TXT	download; read-only
`data/interim/_per_sensor/`	MEDA 5-minute means of each sensor (caches)	Parquet	ingest script
`data/interim/meda.parquet`	MEDA 5-minute table (Table 3.5)	Parquet	ingest script
`data/interim/hourly/`	hourly table of each station (Table 3.6)	Parquet	hourly script
`data/processed/<set>/meda/`	MEDA feature tables of each feature set, with split labels (Chapter 4)	Parquet	feature script
`data/docs/`	README, codebook, terms of use	Markdown, CSV	author
`config/`	parameters of every processing step and experiment	YAML	author
`reports/<run>/`	configuration, log, summaries and results of each run	YAML, TXT, JSON, Parquet	every script

Source: own elaboration.

[16]Rodriguez-Manfredi et al., *op. cit.*, label of the ANCILLARY products.

The derived data are stored as Apache Parquet files, a column-oriented binary format that preserves data types, marks missing values explicitly and compresses well, and they are read and written with the pandas library[17]. The MEDA 5-minute table occupies 22 MB, against 41 GB of source files. The processing keeps its intermediate stages as separate files, so that each stage can be inspected and rerun on its own: the per-sensor caches of 5-minute means, the merged 5-minute table, the hourly tables and the feature tables. Derived files are named in lower-case words separated by underscores, with the site names `meda`, `rems` and `insight` and the names of the two feature sets, `primary_only` and `primary_plus_secondary`. Every run of a script creates its own folder under `reports/`, named by date, time, configuration and code version, for example `20261008_121707_transfer_gbdt_nogit`. The suffix `nogit` records that the project folder was not under version control when the run was made. A superseded intermediate table, built from a preliminary InSight download, is kept under a distinct name and is not used.

The documentation is kept next to the data, in `data/docs/`. It consists of a README that describes each archive, its coverage and its processing, a machine-readable codebook with the content of Tables 3.5 and 3.6, and a copy of the terms of use (Section 3.7). The original documentation of the archives, such as the MEDA readme and release notes, the REMS index and format file and the labels of every product, stays in the raw folders.

3.4 Data cleaning and quality control

Cleaning was limited to the removal of values that the instrument teams mark as invalid or unreliable, plus one artifact found during this work. All rules are applied by the ingest code, and none by hand. Four general rules apply to every station. Missing values of every kind are converted to one representation, the not-a-number value of the numerical libraries, which Parquet stores as null, and no value is imputed at this stage. Time stamps are parsed from text into numbers, while flag strings remain text. Each time step appears only once in the derived tables, and the one-version-per-sol rule above excludes duplicate products. Finally, every removed value is covered by a rule of Table 3.4, whose thresholds are named constants in the code or values in the configuration files.

[17] W. McKinney. “Data Structures for Statistical Computing in Python”. In: *Proceedings of the 9th Python in Science Conference*. 2010, pp. 56–61.

Table 3.4: Quality-control rules applied to the station records

Station	Variable	Rule	Effect
MEDA	all	empty cells and 999999999 set to missing	—
MEDA	humidity	samples before sol 63 removed	humidity from sol 63
MEDA	wind	samples before sol 36 removed	derived wind in 257 sols (36–312)
MEDA	air temperature	mean of the five ATS units; the calibrated product has no quality flags	—
REMS	air temperature	confidence bytes 7 (error within 5 K) and 8 (reliable) equal to 1; samples taken from an off-scale boom (booms differ by more than 50 K) removed	byte 7 is 0 in 9–26 % of the inspected samples; the boom rule removes 0.004 % of samples
REMS	pressure	confidence byte 1 (no shadow effect) equal to 1	—
REMS	wind	confidence bytes 2 (rover still) and 5 (reliable) equal to 1	no sample passes
APSS	air temperature	flag 0 (electronics within temperature range) equal to 1 for each boom; mean of the valid booms; plume and accuracy flags not applied	plume flag reads “affected” in every sample
APSS	wind	flags 0 (electronics) and 1 (1 Hz data) equal to 1	no wind in sols 20–220 (sampled at 0.1–0.5 Hz)
APSS	pressure	no flags; non-numeric values set to missing	—
All	hourly tables	hourly mean kept only with enough samples: MEDA one 5-minute interval, REMS and APSS 60 samples	—

Source: own elaboration based on the documentation of the archives[18].

For MEDA, the rules come from the bundle documentation. Humidity data before sol 63 and wind data before sol 36 precede the first maintenance of these sensors and are not considered reliable, so they are removed[19]. The calibrated air-temperature product carries no quality flags. Air temperature is therefore taken as the mean of all five ATS units, three mounted around the remote sensing mast and two on the sides of the rover body[20], at heights of 1.45 m and 0.84 m[21]. Because the air near the surface is stratified, with a convective period by day and a stable atmosphere with a thermal inversion at night[22], this mean is a fixed mixture of two heights rather than the temperature at one height. It is defined in the same way in every sol, which is what

[18]Rodriguez-Manfredi et al., *op. cit.*; Gomez-Elvira, *op. cit.*; Rodriguez-Manfredi, *op. cit.*

[19]Rodriguez-Manfredi et al., *op. cit.*, `readme.txt` of the bundle.

[20]*Ibid.*, label of the ATS products.

[21]A. Munguira et al. “Near Surface Atmospheric Temperatures at Jezero From Mars 2020 MEDA Measurements”. In: *Journal of Geophysical Research: Planets* 128.3, e2022JE007559 (2023).

[22]Cf. *ibid.*

a forecast of it requires, but it is not directly comparable with the values of the other stations.

For REMS, quality is coded in the confidence-level strings defined in the format file[23]. Air temperature is kept when byte 7 (error within the 5 K operational range) and byte 8 (reliable observation) both equal 1. Byte 8 equaled 1 in every inspected file and removes nothing on its own. Byte 7 equaled 0 in 9–26 % of the samples of the inspected files, mostly between 9 and 15 LMST, yet removing these samples changed the hourly means by a median of 0 K and removed 0.1 % of the hours. Pressure is kept when byte 1 reports no shadow effect on the sensor, and wind when the rover was still (byte 2) and the observation is marked reliable (byte 5).

One artifact passes these flags. REMS measures air temperature on two booms, and its ambient temperature normally follows the warmer of the two. When the booms differ by more than 50 K, the colder one is off scale, at 87–166 K, far below any air temperature measured at Gale. In a few evening hours, mostly after sol 4000, the ambient temperature was nevertheless taken from the off-scale boom while the flags still marked it as reliable. Such samples, 0.004 % of the total, are discarded. Before this rule, 21 hourly values lay between 130 and 180 K, 14 of them in the period later used for testing. After it, the hourly air temperatures at Gale range from 184 to 279 K. A simpler rule that rejects every sample with disagreeing booms would not work, because the booms differ by more than 30 K in 31 % of the inspected samples.

No REMS wind sample passes the wind flags. This agrees with the literature: the wind sensor was damaged at landing, probably by debris[24], and an earlier forecasting study judged its data not accurate enough to use[25]. The transfer experiments of RQ3 therefore cover air temperature and pressure only.

For InSight, the flags are those defined in the labels of the TWINS products[26]. The air temperature of each boom is kept when its electronics were within their temperature range, and the valid booms are averaged. Wind additionally requires data at the full rate of 1 Hz. Two further flags are not applied. The flag that marks samples affected by the thermal plume of the lander reads "affected" in every sample of the inspected files, so applying it would remove the whole record. The accuracy flag, which marks an accuracy worse than 6 K or data at a low rate, is set on 8.6 % of the boom temperatures in thirteen files spread over the mission, and these samples were also kept. The InSight air temperature must therefore be treated as perturbed by the lander. Viúdez-Moreiras et al. state this directly: "The local air temperatures are strongly perturbed by the lander's thermal effects"[27]. The pressure files carry no quality flags, and only non-numeric values are removed from them.

[23] Gomez-Elvira, *op. cit.*, format file `MODRDR6.FMT`.

[24] D. Viúdez-Moreiras et al. "Gale surface wind characterization based on the Mars Science Laboratory REMS dataset. Part I: Wind retrieval and Gale's wind speeds and directions". In: *Icarus* 319 (2019), pp. 909–925.

[25] A. de Cabo García et al. "Prediction of Mars meteorological variables using artificial neural networks". In: *DYNA New Technologies* 9.1 (2022).

[26] Rodriguez-Manfredi, *op. cit.*, labels of the modeled collection.

[27] D. Viúdez-Moreiras et al. "Effects of a Large Dust Storm in the Near-Surface Atmosphere as Measured by InSight in Elysium Planitia, Mars. Comparison With Contemporaneous Measurements by Mars Science Laboratory". In: *Journal of Geophysical Research: Planets* 125.9, e2020JE006493 (2020).

The cleaning keeps some values that look anomalous but are physical. At Gale, the lowest hourly pressure in each block of 500 sols fell steadily, from 689 Pa in sols 1–499 to 642 Pa in sols 4500–4843. Part of this decline reflects the climb of Curiosity up Aeolis Mons, to which the lower mean pressure of later years has been attributed[28]. These values are kept, because they describe the conditions at the rover. Other unusual values, such as the extreme tail of the upward longwave flux measured by MEDA and its raw relative humidity, are examined in Chapter 4.

3.5 Codebook of the derived tables

Two kinds of derived tables feed the models. The MEDA 5-minute table (Table 3.5) has 255,132 rows, one per 5-minute interval of the spacecraft clock in which at least one sensor recorded data, and 19 columns. It serves the single-site experiments of RQ1 and RQ2. The hourly tables (Table 3.6) share one schema for all three stations, with one row per sol and LMST hour. They hold 33,411 rows for MEDA, 110,559 for REMS and 18,785 for InSight and serve the transfer experiments of RQ3. The aggregation of the samples into these grids is described in Chapter 4. In both tables, a missing value is stored as null and never as a numerical code, and the typical range is the interval between the 1st and the 99th percentile of the values.

Several entries of the codebook need attention. The variable `solar_longitude` keeps the misleading name of its source column but contains the solar azimuth. The seasonal L_s is computed separately, in the hourly tables from UTC and in the MEDA features approximately, from the sol number (Chapter 4). The MEDA `sol` column is taken from the pressure product, so it is missing in the 5.1 % of intervals without a pressure sample, which are therefore left out of the data splits, and it takes a half-integer value in 600 intervals that span two daily files. The raw relative humidity `RH` refers to the temperature of the sensor head rather than of the air and reaches physically meaningless values, so it is kept only as a diagnostic, and the volume mixing ratio `vmr_h2o` serves as the humidity variable. Finally, the typical ranges show how much the sites differ. The InSight air temperature reaches lower values than at the two crater sites (1st percentile 174.5 K against 192–193 K), and the median hourly pressure at Gale (825 Pa) is about 100 Pa higher than at Jezero (715 Pa) and at Elysium Planitia (729 Pa).

3.6 Forecast targets and labels

The forecasting task is supervised, but its labels are not annotations. The target of each example is a later value of the same series: the air temperature, pressure or wind speed 1, 6, 24 or 48 hours after the time at which the forecast is issued or, in the transfer experiments, the change of that value over the horizon. The labels are therefore produced automatically from the measurements, without human judgment, and annotation guidelines or agreement between annotators do not apply. Examples whose target value is missing are removed rather than imputed, so that every

[28] Cf. D. Viúdez-Moreiras et al. “Effects of the MY34/2018 Global Dust Storm as Measured by MSL REMS in Gale Crater”. In: *Journal of Geophysical Research: Planets* 124.7 (2019), pp. 1899–1912.

Table 3.5: Codebook of the MEDA 5-minute table

Column	Type	Unit	Meaning and special values	Typical range
`sclk_bin`	integer	300 s	index of the 5-minute interval, SCLK divided by 300 and rounded down; row key	—
`sol`	float	sol	sol of the MEDA pressure file; missing in 5.1 % of rows (no pressure sample), half-integer in 600 rows that span two files	1–1619[a]
`lmst_hour`	float	h	LMST as a fractional hour	0–24[a]
`T_air`	float	K	air temperature, mean of the five ATS units	192.7–263.4
`pressure`	float	Pa	surface pressure	597.7–800.4
`wind_speed`	float	$\mathrm{m\,s^{-1}}$	horizontal wind speed; present only in sols 36–312	0.5–8.7
`wind_speed_z`	float	$\mathrm{m\,s^{-1}}$	vertical wind speed; empty in the release used	—
`wind_dir`	float	degree	wind direction; present only in sols 36–312	38–331
`RH`	float	%	relative humidity with respect to the temperature of the sensor head; diagnostic only	0–41
`T_humidity_local`	float	K	temperature of the humidity sensor head	194.1–265.0
`vmr_h2o`	float	ppm	volume mixing ratio of water vapor; the humidity variable	11.3–111.2
`LW_down`	float	$\mathrm{W\,m^{-2}}$	downward longwave irradiance	19.9–59.3
`LW_up`	float	$\mathrm{W\,m^{-2}}$	upward longwave irradiance from the ground; rare extreme values	63.0–388.1
`RDS_TOP_avg`	float	$\mathrm{W\,m^{-2}}$	mean of the eight photodiodes on the top of the RDS (normal gain)	0.0–46.2
`solar_longitude`	float	degree	solar azimuth in the frame of the rover, not L_s (name of the source column kept)	−164–167
`solar_zenith`	float	degree	solar zenith angle	5.8–171.2
`ROVER_PITCH`, `ROVER_ROLL`, `ROVER_YAW`	float	degree	attitude of the rover; diagnostic only	pitch and roll within ±18

[a] minimum to maximum; elsewhere the 1st to 99th percentile.

Source: own elaboration.

Table 3.6: Codebook of the hourly tables of the three stations

				Typical range		
Column	Type	Unit	Meaning	MEDA	REMS	APSS
`site`	categorical	—	station: `meda`, `rems` or `insight`	—	—	—
`sol`	integer	sol	sol of the mission	1–1619[a]	1–4843[a]	4–1333[a]
`hour`	integer	h	LMST hour, 0 to 23	—	—	—
`t`	integer	h	Mars-hour index, 24× sol + hour; row key with `site`	—	—	—
`utc`	date-time	UTC	median time of the samples in the hour	—	—	—
`ls`	float	degree	solar longitude L_s, computed from `utc`	0–360[a]	0–360[a]	0–360[a]
`T_air`	float	K	mean air temperature	192.4–261.7	193.2–266.2	174.5–258.1
`pressure`	float	Pa	mean pressure	598.9–801.3	670.6–938.8	609.0–800.6
`wind_speed`	float	$\mathrm{m\,s^{-1}}$	mean horizontal wind speed	0.6–8.1	absent	0.8–12.2
`wind_u`	float	$\mathrm{m\,s^{-1}}$	mean of speed × cos(direction)	−4.9–3.2	absent	−8.9–5.4
`wind_v`	float	$\mathrm{m\,s^{-1}}$	mean of speed × sin(direction)	−4.5–7.8	absent	−9.2–9.4
`n_T_air`	integer	count	valid samples behind `T_air`[b]	1–13	115–3,619	370–3,700
`n_pressure`	integer	count	valid samples behind `pressure`[b]	1–13	128–3,699	7,398–36,991
`n_wind_speed`	integer	count	valid samples behind `wind_speed`[b]	0–13	absent	800–3,700

[a] minimum to maximum; elsewhere the 1st to 99th percentile. [b] for MEDA, 5-minute intervals; for REMS and APSS, raw samples. A mean with fewer than the minimum number of samples (Table 3.4) is stored as null.

Source: own elaboration.

score is computed against an actual measurement. The exact construction of the targets and of the chronological training, validation and test periods is given in Chapter 6.

3.7 Licensing, ethics and attribution

The archives contain no license file. According to the PDS, "Data sets archived in the PDS are in the public domain". They are classified as publicly available technology that may be exported as open literature, and they "should be cited appropriately" when they are used in publications[29]. The PDS recommends citing the most granular product that has a DOI[30]. Each archive is therefore cited by the DOI of its bundle or data set, together with the sols and processing levels used (Table 3.1), which also credits the instrument teams that produced the data. These terms are saved, with their sources and the date of access, in `data/docs/terms_of_use.md`.

The data raise no privacy or ethical issues. They are physical measurements made by instruments on Mars and contain no personal data and no information about human or animal subjects. All figures and tables of the thesis were produced by the author from these data or from the cited literature, and tables that summarize other works cite them in their source lines.

3.8 Reproducibility and data limitations

The path from the archives to the tables of this chapter is scripted. A download script mirrors the archive folders, an ingest script builds the MEDA 5-minute table, an hourly script builds the hourly table of each station and a feature script builds the MEDA feature tables. Every parameter of these steps, from the sol range and the excluded sensor periods to the length of the aggregation interval and the minimum number of samples per hour, is set in a configuration file or, for the flag positions and the boom threshold, in a named constant of the code. Each run writes its resolved configuration, a log and a summary to its own report folder, and no data file was edited by hand. Rebuilding the MEDA 5-minute table takes about 75 minutes from the source files and about 30 seconds from the per-sensor caches.

The data also limit what the later chapters can show:

- *Representativeness.* An hourly mean rests on a median of 293 samples, about five minutes of data, at Gale. At Jezero, 28 % of the hourly values rest on at most ten minutes and 38 % on at least 55 minutes. At Elysium Planitia, the median is 3,699 samples, a full hour.
- *Sensor placement.* The MEDA air temperature mixes two heights, and the InSight air temperature is perturbed by the lander and includes samples of reduced accuracy.
- *Wind.* Derived MEDA wind covers only sols 36–312, so wind is evaluated within that window, and no REMS wind passes the quality control, so the transfer experiments cover

[29] PDS Geosciences Node. *Frequently Asked Questions*. URL: `https://pds-geosciences.wustl.edu/faq.html` (accessed 2026-10-18).

[30] NASA Planetary Data System. *Citing PDS4 Data*. URL: `https://pds.nasa.gov/datastandards/citing/` (accessed 2026-10-18).

air temperature and pressure only.

- *Late InSight coverage.* The usable InSight record spans about sols 4–870, and six late pressure files are missing.
- *Version control.* The project folder was not under version control, so results are tied to the configuration and log of each run rather than to a code revision.

Bibliography

Allison, M. and McEwen, M. "A Post-Pathfinder Evaluation of Areocentric Solar Coordinates with Improved Timing Recipes for Mars Seasonal/Diurnal Climate Studies". In: *Planetary and Space Science* 48.2–3 (Feb. 2000), pp. 215–235. DOI: 10.1016/S0032-0633(99)00092-6.

Banfield, D. *APSS PS Data*. NASA Planetary Data System, 2019. DOI: 10.17189/1518939. URL: https://pds.nasa.gov/ds-view/pds/viewBundle.jsp?identifier=urn:nasa:pds:insight_ps (accessed 2026-10-18).

Banfield, D. et al. "The atmosphere of Mars as observed by InSight". In: *Nature Geoscience* 13.3 (2020), pp. 190–198. DOI: 10.1038/s41561-020-0534-0.

de Cabo García, A. et al. "Prediction of Mars meteorological variables using artificial neural networks". In: *DYNA New Technologies* 9.1 (2022). DOI: 10.6036/NT10369.

Gomez-Elvira, J. *Mars Science Laboratory Rover Environmental Monitoring Station RDR Data V1.0*. MSL-M-REMS-5-MODRDR-V1.0. NASA Planetary Data System, 2013. DOI: 10.17189/ghbc-qr41. URL: https://pds.nasa.gov/ds-view/pds/viewDataset.jsp?dsid=MSL-M-REMS-5-MODRDR-V1.0 (accessed 2026-10-18).

Martínez, G. M. et al. "The Modern Near-Surface Martian Climate: A Review of In-situ Meteorological Data from Viking to Curiosity". In: *Space Science Reviews* 212.1–2 (2017), pp. 295–338. DOI: 10.1007/s11214-017-0360-x.

McKinney, W. "Data Structures for Statistical Computing in Python". In: *Proceedings of the 9th Python in Science Conference*. 2010, pp. 56–61. DOI: 10.25080/Majora-92bf1922-00a.

Munguira, A. et al. "Near Surface Atmospheric Temperatures at Jezero From Mars 2020 MEDA Measurements". In: *Journal of Geophysical Research: Planets* 128.3, e2022JE007559 (2023). DOI: 10.1029/2022JE007559.

NASA Planetary Data System. *Citing PDS4 Data*. URL: https://pds.nasa.gov/datastandards/citing/ (accessed 2026-10-18).

PDS Geosciences Node. *Frequently Asked Questions*. URL: https://pds-geosciences.wustl.edu/faq.html (accessed 2026-10-18).

Rodriguez-Manfredi, J. A. *APSS TWINS Data*. NASA Planetary Data System, 2019. DOI: 10.17189/1518950. URL: https://pds.nasa.gov/ds-view/pds/viewBundle.jsp?identifier=urn:nasa:pds:insight_twins (accessed 2026-10-18).

Rodriguez-Manfredi, J. A. et al. "The diverse meteorology of Jezero crater over the first 250 sols of Perseverance on Mars". In: *Nature Geoscience* 16 (2023), pp. 19–28. DOI: 10.1038/s41561-022-01084-0.

Rodriguez-Manfredi, J. A. et al. *The Mars Environmental Dynamics Analyzer, MEDA*. Data bundles of the Mars 2020 mission. NASA Planetary Data System, Atmospheres Node. 2021. URL: https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/PERSEVERANCE/meda.html (accessed 2026-10-07). DOI: 10.17189/1522849.

Viúdez-Moreiras, D. et al. “Effects of a Large Dust Storm in the Near-Surface Atmosphere as Measured by InSight in Elysium Planitia, Mars. Comparison With Contemporaneous Measurements by Mars Science Laboratory”. In: *Journal of Geophysical Research: Planets* 125.9, e2020JE006493 (2020). DOI: 10.1029/2020JE006493.

Viúdez-Moreiras, D. et al. “Effects of the MY34/2018 Global Dust Storm as Measured by MSL REMS in Gale Crater”. In: *Journal of Geophysical Research: Planets* 124.7 (2019), pp. 1899–1912. DOI: 10.1029/2019JE005985.

Viúdez-Moreiras, D. et al. “Gale surface wind characterization based on the Mars Science Laboratory REMS dataset. Part I: Wind retrieval and Gale’s wind speeds and directions”. In: *Icarus* 319 (2019), pp. 909–925. DOI: 10.1016/j.icarus.2018.10.011.